

PROFESSIONAL SYLLABUS

Comprehensive Certification Program

Vulnerability Assessment & Penetration Testing — a hands-on, industry-aligned curriculum covering reconnaissance, exploitation, web & mobile pentesting, post-exploitation, professional reporting, and career readiness.

MODULES

13 Modules

LEVEL

**Beginner →
Advanced**

FORMAT

Theory + Labs

OUTCOME

Industry-Ready



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Syllabus Index

A structured pathway from foundational cybersecurity concepts to advanced offensive security operations, professional reporting, and career launch.

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Course Modules

MODULE 01 Foundations of Cybersecurity and VAPT

- 1.1 Introduction to Cybersecurity Concepts
- 1.2 What is VAPT? Definitions and Distinctions
- 1.3 Types of Penetration Testing
- 1.4 Legal and Ethical Frameworks
- 1.5 Industry Standards and Methodologies (PTES, OWASP, OSSTMM, NIST)
- 1.6 Setting Up a Safe Lab Environment

MODULE 02 Reconnaissance and Information Gathering

- 2.1 Passive Reconnaissance Techniques
- 2.2 Active Reconnaissance Techniques
- 2.3 OSINT (Open Source Intelligence) Tools and Methods
- 2.4 DNS Enumeration and Analysis
- 2.5 Network Footprinting
- 2.6 Social Engineering as a Reconnaissance Vector

MODULE 03 Scanning and Enumeration

- 3.1 Network Scanning Fundamentals
- 3.2 Port Scanning Techniques (TCP, UDP, SYN)
- 3.3 Service and Version Detection
- 3.4 OS Fingerprinting
- 3.5 Vulnerability Scanning
- 3.6 Enumeration of Network Services (SMB, SNMP, LDAP, NFS)
- 3.7 Web Application Enumeration

MODULE 04 Vulnerability Assessment

- 4.1 Understanding Vulnerability Classes and Categories
- 4.2 CVE, CVSS, and NVD — Scoring and Databases
- 4.3 Automated Vulnerability Scanners (Nessus, OpenVAS, Qualys)
- 4.4 Manual Vulnerability Identification Techniques
- 4.5 Risk Rating and Prioritization
- 4.6 Vulnerability Assessment Reporting

MODULE 05 Exploitation — System and Network

- 5.1 Exploitation Concepts and Frameworks
- 5.2 Metasploit Framework — Architecture and Usage
- 5.3 Exploiting Common Network Vulnerabilities
- 5.4 Password Attacks (Brute Force, Dictionary, Hash Cracking)
- 5.5 Man-in-the-Middle (MitM) Attacks
- 5.6 Buffer Overflow Concepts
- 5.7 Exploiting Misconfigurations

MODULE 06 Web Application Penetration Testing

- 6.1 Web Application Architecture and HTTP Fundamentals
- 6.2 OWASP Top 10 (2021) — In-Depth Coverage
- 6.3 SQL Injection (SQLi)
- 6.4 Cross-Site Scripting (XSS)
- 6.5 Cross-Site Request Forgery (CSRF)
- 6.6 Authentication and Session Management Flaws
- 6.7 API Security Testing
- 6.8 Business Logic Vulnerabilities
- 6.9 Web Application Firewall (WAF) Bypass Techniques
- 6.10 Tools: Burp Suite, OWASP ZAP, Nikto

MODULE 07**Mobile Application Penetration Testing**

- 7.1 Mobile Security Fundamentals (Android and iOS)
- 7.2 Android Application Testing (APK Analysis, Emulators)
- 7.3 iOS Application Testing (IPA Analysis, Jailbreaking Concepts)
- 7.4 OWASP Mobile Top 10
- 7.5 Intercepting Mobile Traffic
- 7.6 Mobile-Specific Tools (MobSF, Frida, Apktool)

MODULE 08**Network Penetration Testing**

- 8.1 Internal vs. External Network Testing
- 8.2 Active Directory Attacks and Enumeration
- 8.3 Wireless Network Security Testing (WPA2, WPA3, WPS)
- 8.4 VPN and Firewall Testing
- 8.5 Network Protocol Exploitation
- 8.6 Cloud Infrastructure Testing Fundamentals

MODULE 09**Post-Exploitation**

- 9.1 Post-Exploitation Goals and Concepts
- 9.2 Privilege Escalation (Windows and Linux)
- 9.3 Lateral Movement Techniques
- 9.4 Persistence Mechanisms
- 9.5 Data Exfiltration Methods
- 9.6 Covering Tracks and Anti-Forensics
- 9.7 Pivoting and Tunneling

MODULE 10 Social Engineering

- **10.1** Social Engineering Concepts and Psychology
- **10.2** Phishing, Spear Phishing, and Whaling
- **10.3** Vishing and Smishing
- **10.4** Physical Security Testing (Tailgating, Impersonation)
- **10.5** Pretexting and Campaign Design
- **10.6** Tools: GoPhish, SET (Social Engineering Toolkit)

MODULE 11 Reporting and Documentation

- **11.1** Importance of VAPT Reporting
- **11.2** Anatomy of a Professional Pentest Report
- **11.3** Executive Summary vs. Technical Report
- **11.4** Evidence Collection and Chain of Custody
- **11.5** Remediation Recommendations
- **11.6** Debriefing and Communication with Stakeholders
- **11.7** Sample Report Templates

MODULE 12 Specialized Testing Domains

- **12.1** IoT Security Testing
- **12.2** Cloud Security Testing (AWS, Azure, GCP)
- **12.3** Industrial Control Systems (ICS/SCADA) Security
- **12.4** Red Team vs. Blue Team vs. Purple Team Operations
- **12.5** Threat Intelligence Integration

MODULE 13**Career Development and Certifications**

- **13.1** VAPT Career Pathways
- **13.2** Key Certifications (CEH, OSCP, GPEN, eJPT, BSCP)
- **13.3** Building a Portfolio and Lab Environment
- **13.4** Bug Bounty Programs
- **13.5** Staying Current — CTFs, Communities, and Research

Tools Covered

Industry-standard tools used by professional penetration testers across the engagement lifecycle.

Reconnaissance & Scanning

- Nmap, Masscan
- theHarvester, Maltego, Recon-ng
- Shodan, Censys
- Amass, Sublist3r, Dnsrecon

Vulnerability Assessment

- Nessus
- OpenVAS / Greenbone
- Qualys
- Nuclei

Exploitation

- Metasploit Framework
- Hydra, John the Ripper, Hashcat
- Responder, Impacket Suite
- BloodHound, CrackMapExec

Web Application Testing

- Burp Suite (Community / Pro)
- OWASP ZAP
- Nikto, SQLMap, ffuf, gobuster
- Postman (API testing)

Mobile Application Testing

- MobSF (Mobile Security Framework)
- Frida, Objection
- Apktool, jadx
- Android Studio Emulator, Genymotion

Wireless & Network

- Aircrack-ng Suite
- Wireshark, tcpdump
- Bettercap, Ettercap
- Kismet

Social Engineering

- GoPhish
- SET (Social Engineering Toolkit)
- Evilginx2
- King Phisher

Platforms & Labs

- Kali Linux, Parrot OS
- TryHackMe, Hack The Box
- PortSwigger Web Security Academy
- VulnHub, DVWA, OWASP Juice Shop

Certifications Awarded

Upon successful completion of the program and assessment, students receive:

Course Completion Certificate

Internship Certificate

Live Project Certificate

VAPT Practitioner Badge

Career Opportunities

Roles you can pursue after completing the Comprehensive VAPT Certification Program:

Offensive Security Roles

- Penetration Tester (Web / Network / Mobile)
- Red Team Operator
- Application Security Engineer
- Bug Bounty Hunter

Defensive & Hybrid Roles

- Vulnerability Assessment Analyst
- SOC Analyst (Tier 1 / Tier 2)
- Threat Intelligence Analyst
- Purple Team Engineer

Consulting & Audit Roles

- Cybersecurity Consultant
- VAPT Consultant
- Security Auditor
- GRC (Governance, Risk & Compliance) Analyst

Specialized & Research Roles

- Cloud Security Engineer
- IoT / ICS Security Researcher
- Security Researcher
- Independent Security Consultant / Freelancer

Program Promise: This syllabus is mapped to globally recognized frameworks (PTES, OWASP, OSSTMM, NIST SP 800-115) and prepares learners for industry certifications including **CEH**, **eJPT**, **OSCP**, **GPEN**, and **BSCP**. Every module is reinforced with hands-on labs, real-world case studies, and a capstone reporting project.

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Ready to launch your ?

Join Obscyratech's Comprehensive VAPT Certification Program and gain the hands-on skills, professional reporting capabilities, and industry knowledge required to thrive as a penetration tester or security consultant.

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LOCATION

Mysore, Karnataka, India

PROGRAM

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Certification Program**